CEG5101 Modern Computer Networking

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Laboratory 0

Install Ubuntu on Oracle VirtualBox

LEARNING OBJECTIVES

Upon successful completion of this laboratory, the students will be able to learn:

i. To download and install Ubuntu on VirtualBox

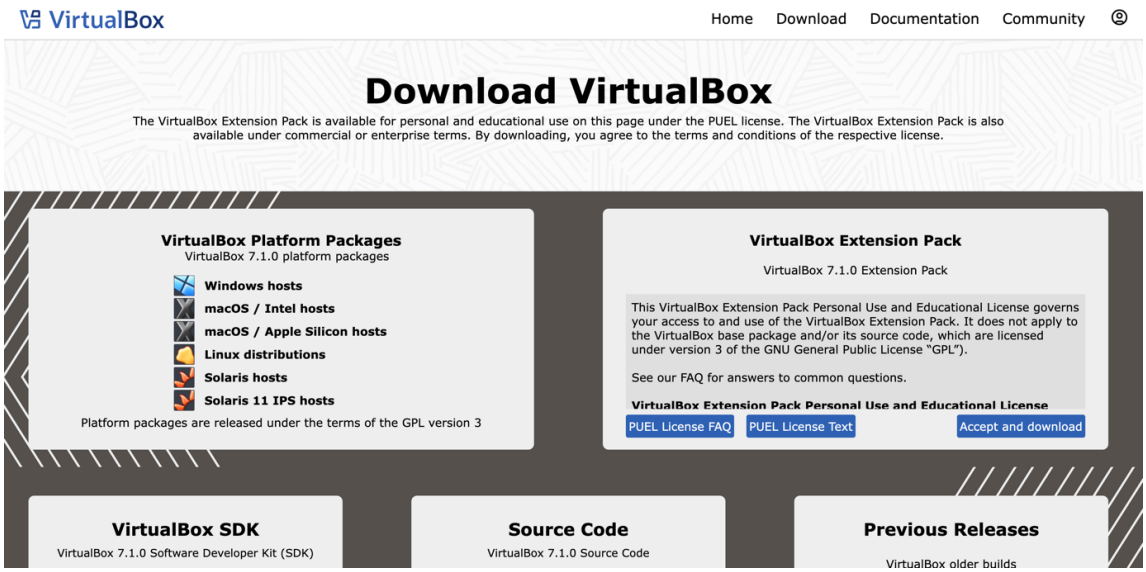
EQUIPMENT

i. *Hardware* – Laptop/Computer with internet access

DISCUSSION

1. Oracle VirtualBox

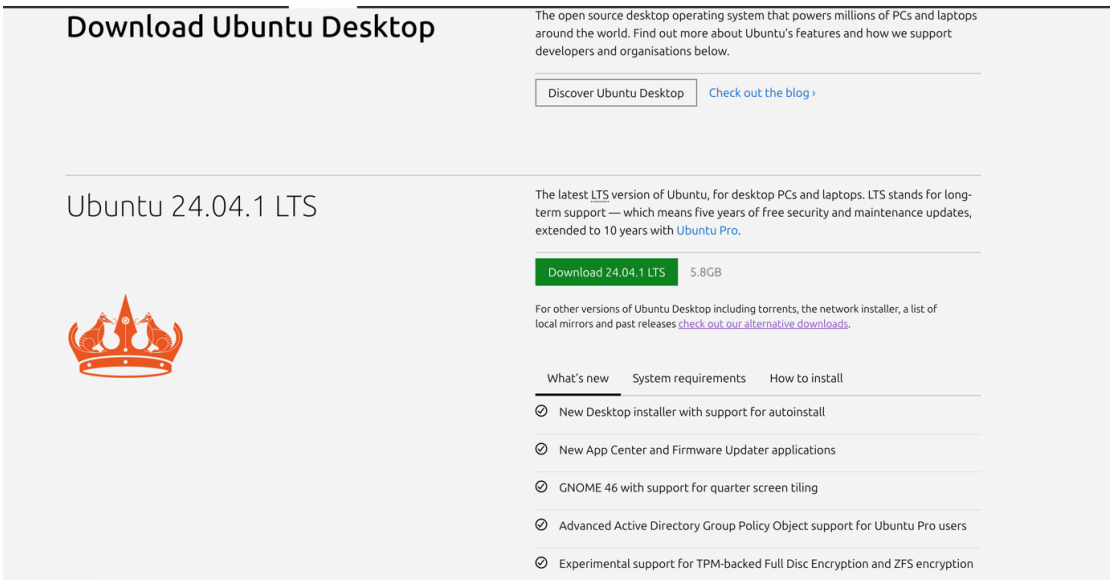
a. Go to https://www.virtualbox.org/wiki/Downloads
b. Download VirtualBox latest version (currently 7.2.6) platform packages for WIndows or OS X (MAC users)
c. Run the executable file. Continue with the installation of VirtualBox with the defaults. This will open VirtualBox at the end of the installation.

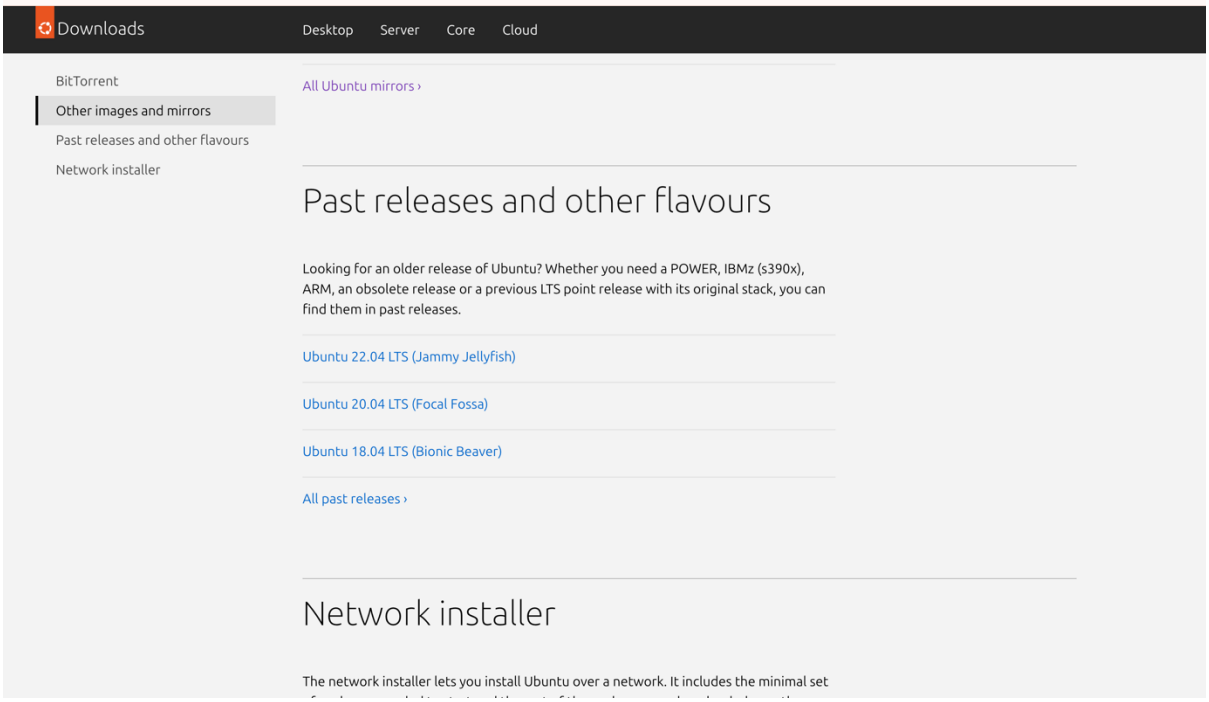


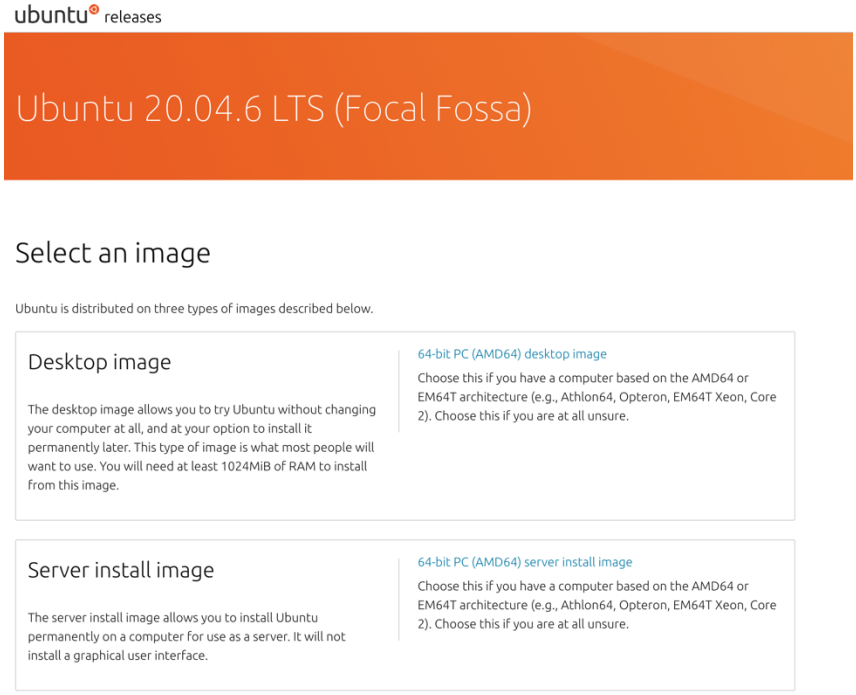
2. Ubuntu Desktop

Download the version **20.04** of Ubuntu from https://ubuntu.com/download/desktop
Ubuntu 20.*04 (python 3.8, recommend for Lab 3 Ryu controller)*

https://releases.ubuntu.com/20.04.6/?_gl=1*byw9gv*_gcl_au*MTQ3NzAzNDU1NS4xNzI3NDMyMjAx&_ga=2.189861864.216104226.1728103719-756818911.1727432197

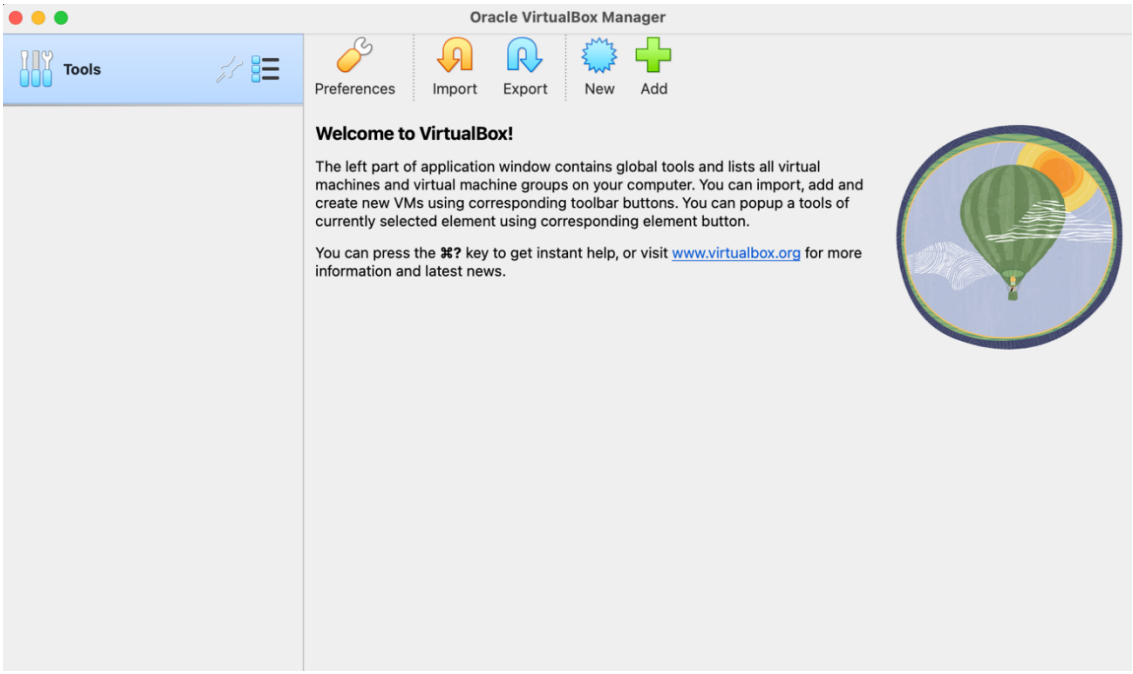






3. Installation screenshots

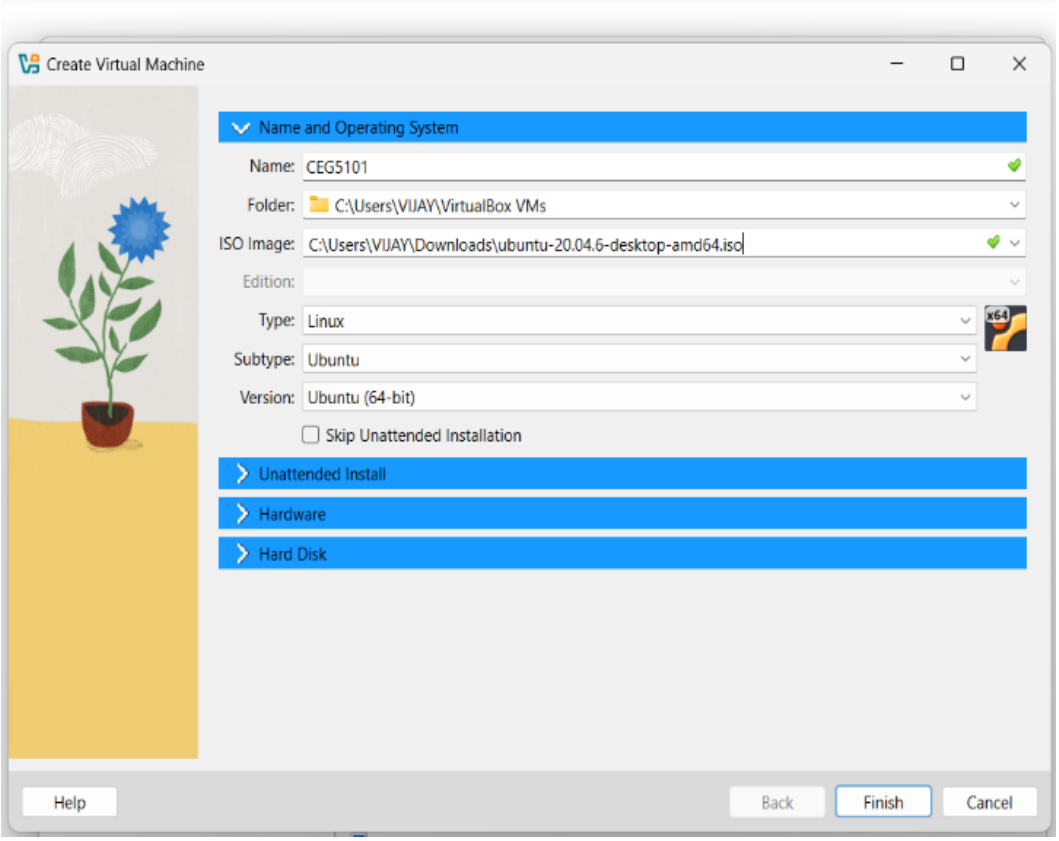
Open Oracle VirtualBox and Click “New”



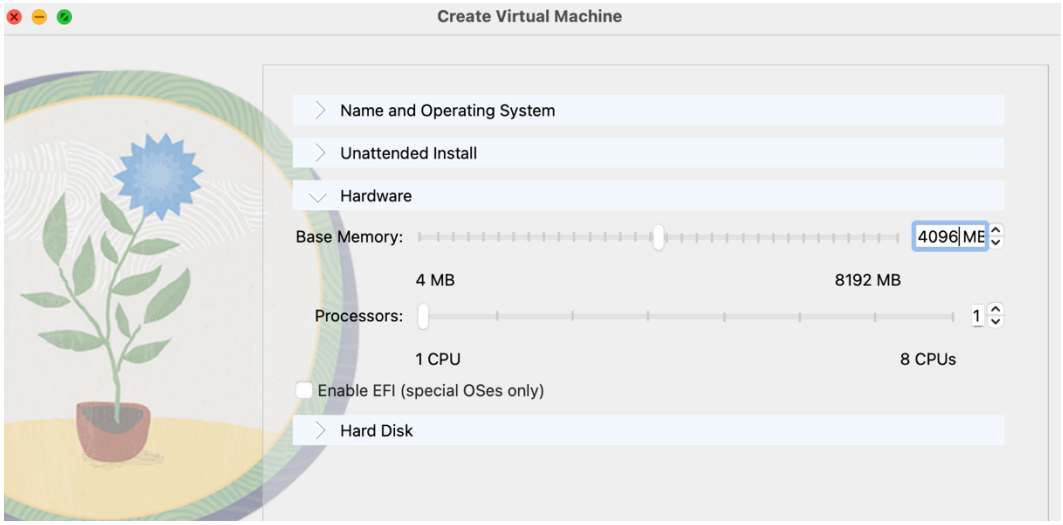
Input:

1. The name of the VM
2. The folder you want to save the VM
3. The ISO image downloaded from the website (Ubuntu file)

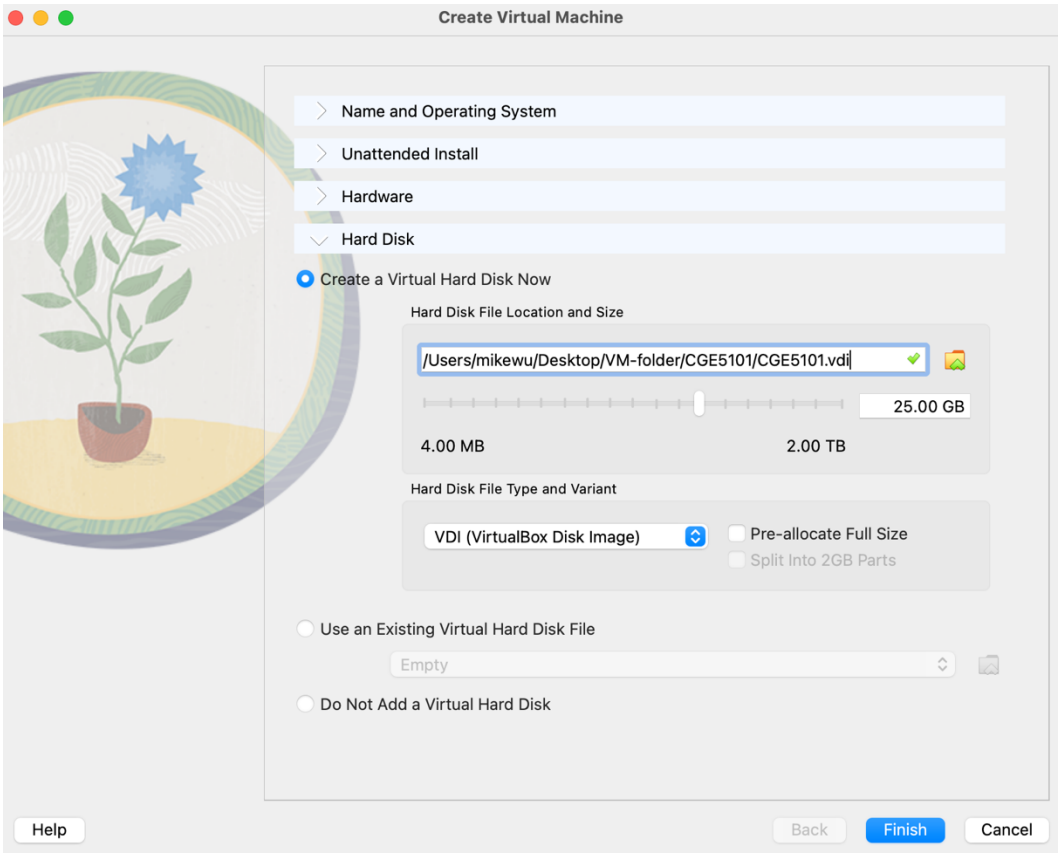
4. You can choose “skip the Unattended Installation” to avoid downloading unnecessary materials



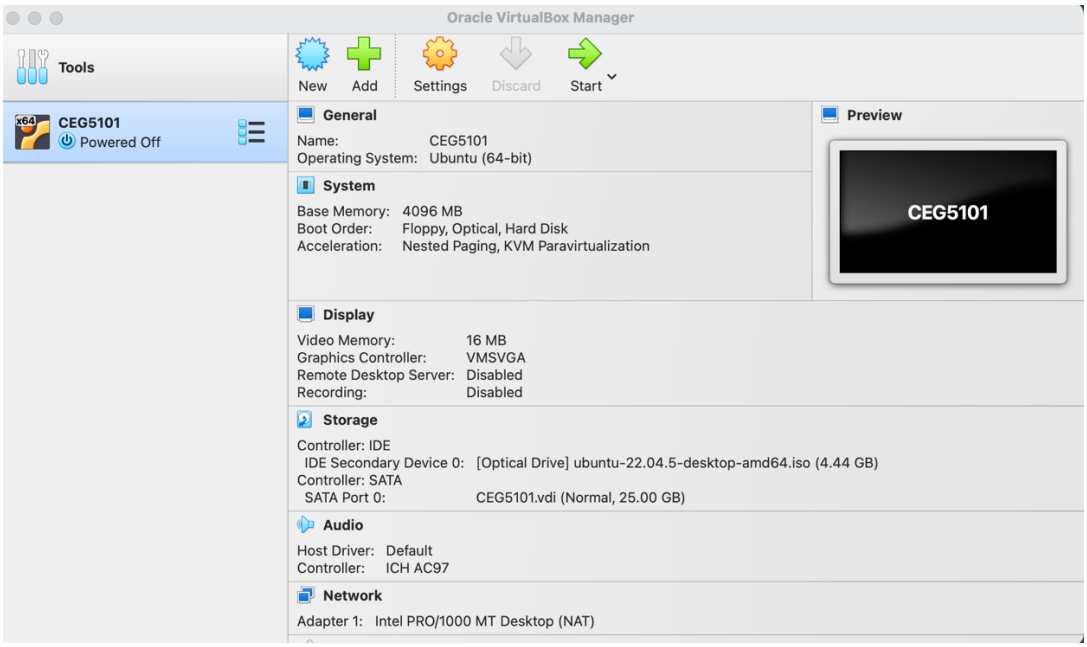
Keep memory around 4GB ~ 4096 (recommended)



Hard Disk size can keep to 25GB



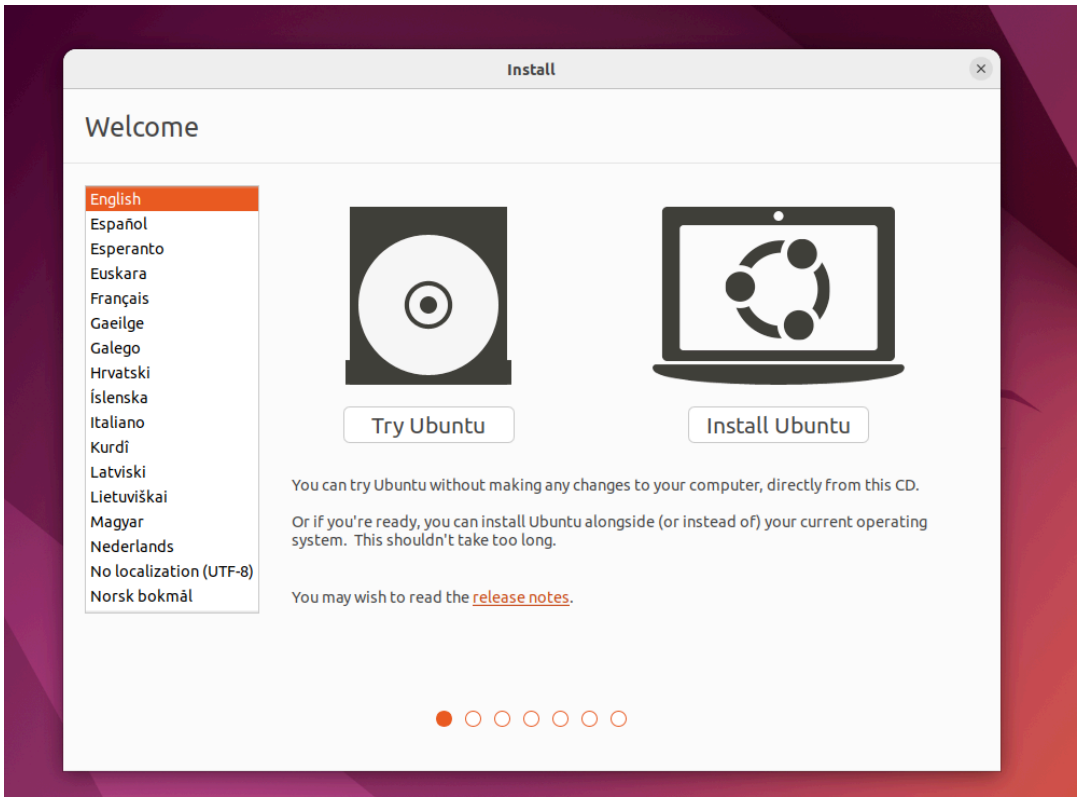
Click “Finish” and Click “start” installation of ubuntu starts form there



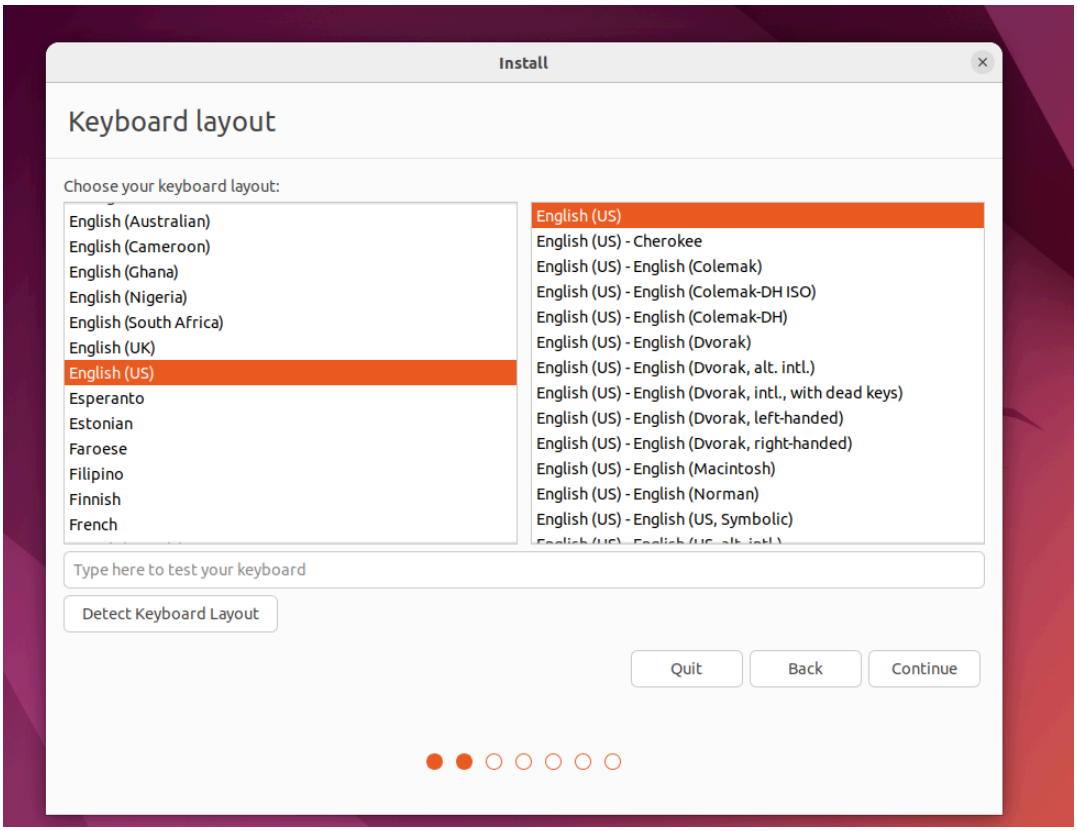
Choose try or install Ubuntu



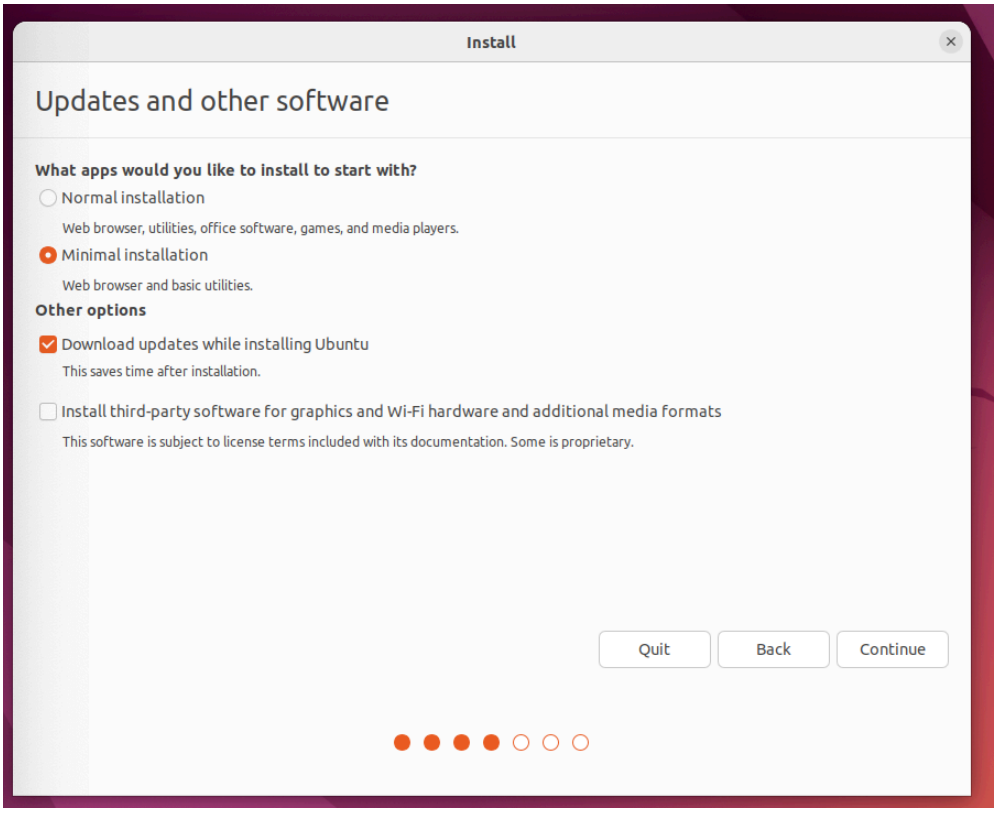
Choose Install Ubuntu



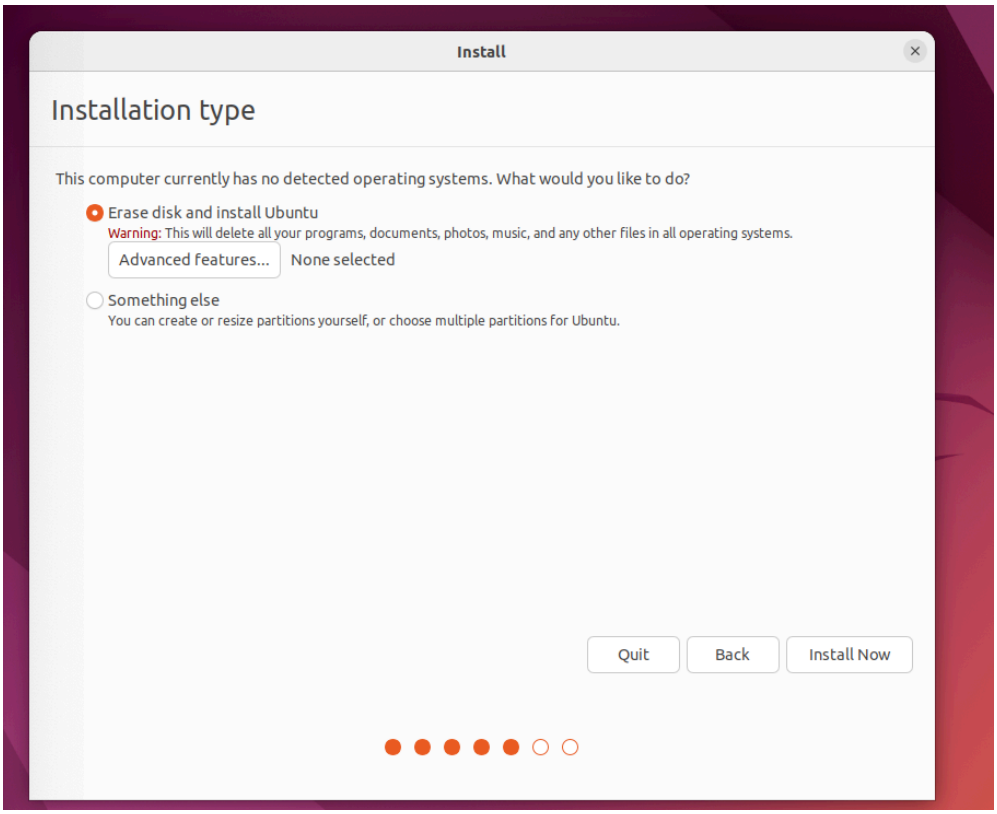
Choose Keyboard layout



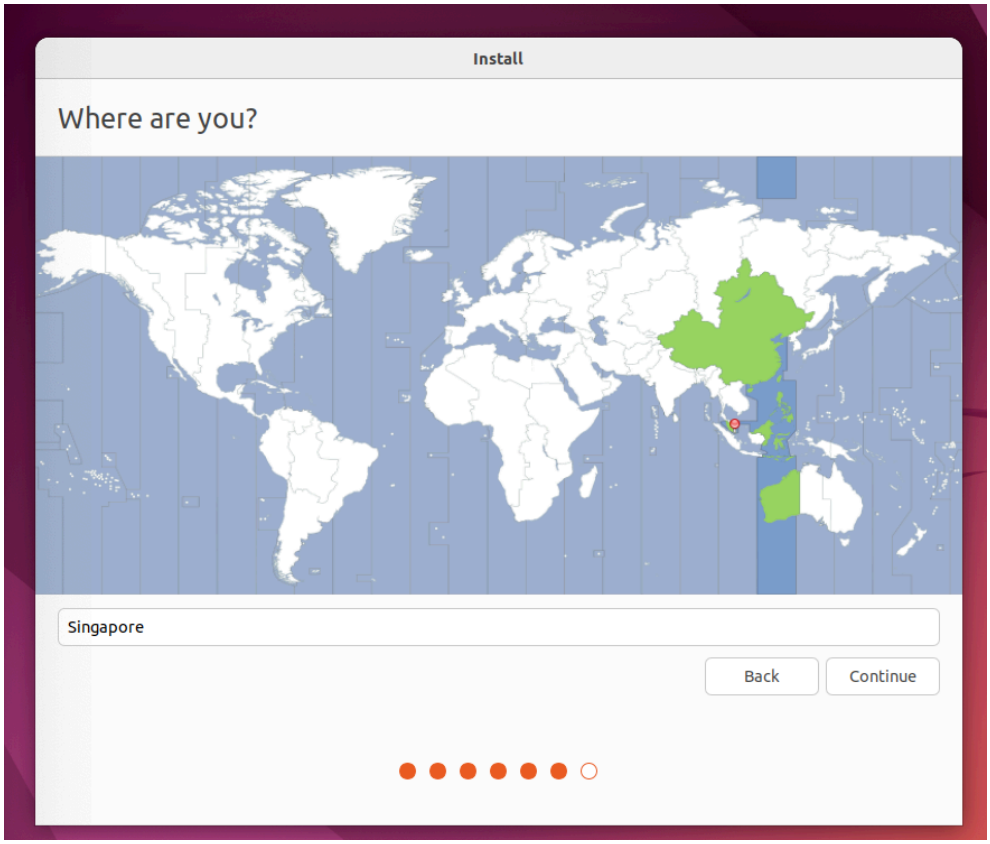
Choose “Minimal Installation”. It is enough for our system. You can also choose “Normal installation” for a more powerful system.



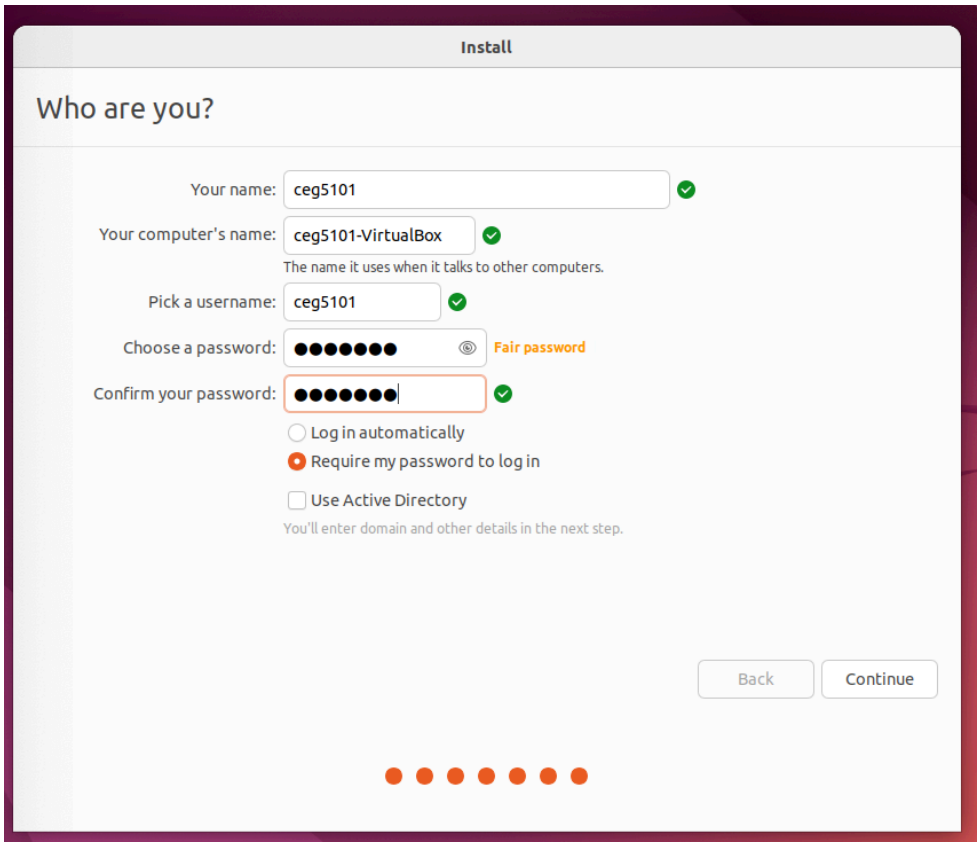
Erase and Install Ubuntu, ignore warning and continue



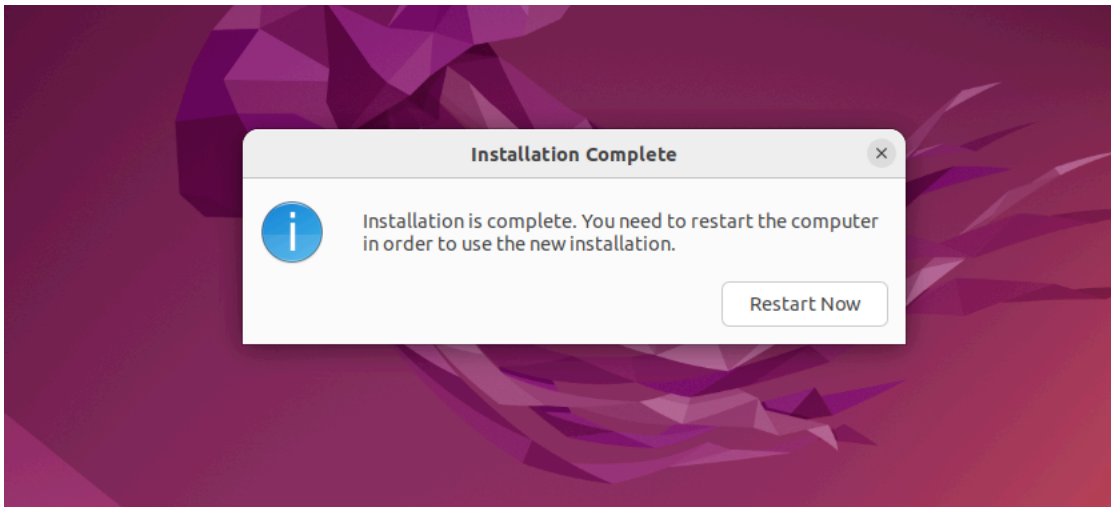
Location: “Singapore”



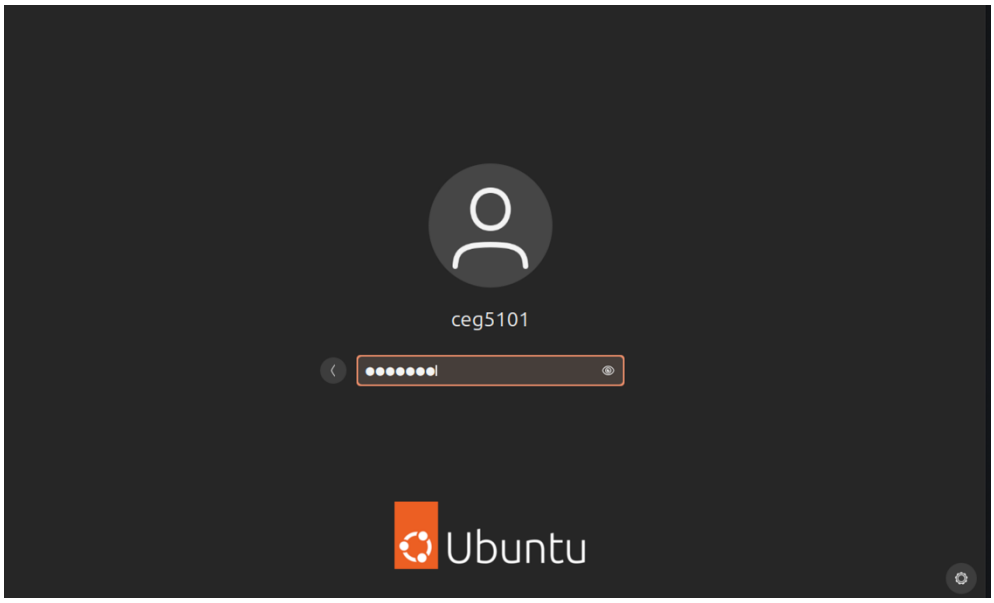
User name and password settings



Waiting for the installation to be finished



Done with the installation of ubuntu in oracle virtual box
You can restart to use the system



ADITIONAL TIPS:

A few useful tips to make your life easier

- Adjust the window size (choose what you want)
  a) View → Auto-resize Guest Display
  b) Virtual_screen1 → 200%
- Copy bidirectional
  c) First, go to settings (make sure you power off the VM)
    i. general –>advanced → sharedclipboard → bidirectional
  d) Open VM and terminal
    i. $ sudo apt update
    ii. $ sudo apt upgrade
  e) Mount the VirtualBox Guest Additions CD Image
    i. In VirtualBox, go to the "Devices" menu at the top of the virtual machine windSow and select "Insert Guest Additions CD Image." This action virtually mounts the Guest Additions CD image into your virtual machine.
  f) Install VirtualBox Guest Additions
    i. cd /media/(your host name, i.e., ceg5101)/VBox_GAs_7.0.10/
    ii. sudo ./VBoxLinuxAdditions.run
    iii. sudo reboot

Then you can paste the code. (It should work, if it does not, try to figure it out by yourself.)

- Install Git
  g) sudo apt install git
  h) sudo apt update

- check python version
  i) python3 --version
  j) python --version